Niccolo' Fineschi

Biomedical AI Researcher

Information

niccofine.github.io | niccofine@gmail.com | 0009-0006-9613-3159 | niccofine | niccofine

EDUCATION

MSc in Biomedical Engineering
eHealth specialization - Grade 106/110
Mar 2024 - Sep 2026
Politecnico di Torino (Italy)

BSc in Biomedical Engineering
Generic specialization
Sep 2019 - Mar 2024
Politecnico di Torino (Italy)

ABOUT

Experience in **representation** and **longitudinal** disease modeling. Recent research focus on CLIP-based visual representations and **continuous-time** forecasting from irregularly sampled observations.

EXPERIENCE

Guest Researcher
Feb 2026 - Aug 2026
University of Osaka (Japan)

Visiting research for MSc thesis at *Intelligence and Sensing Lab* (ISLab) in collaboration with *WPI Premium Research Institute for Human Metaverse Medicine* (WPI-PRIMe).
Supervision by *Panic J.* (University of Osaka), *Rosati S.* and *Balestra G.* (Politecnico di Torino).

***MSc thesis:* Through the eyes of a Vision-Language Model**
Age-related Macular Degeneration (AMD) analysis using fundus images

- **Latent space analysis** using visual encoders from **Vision-Language Models**.
  Probed multiple frozen *CLIP*-based encoders on AREDS retinal images. *RET-CLIP* yielded the most AMD-informative representations. Linked AMD phenotypes to latent dimensions via statistical analysis.
- **Classification and prediction** of AMD status.
  Implemented multiple current-status and future-progression *Machine Learning* AMD classifiers. Probed with *SHAP* explainability and attention-feature maps.
- **Forecasting** disease trajectories with **Neural Dynamical Systems** (NDS).
  Built and benchmarked multiple *Neural ODE* variants for continuous-time forecasting from irregularly sampled visits. *Liquid Neural Networks* performed best (MAE $= 0.78$ on the AREDS $0-5$ severity scale, $1-13$ year horizons). With uncertainty quantification and attention maps for interpretability.

PUBLICATIONS

Proceedings

- **Fineschi N.**, Panic J., Watanabe A., Hayashi H., Nishimura K., Maeno Y., Rosati S., Balestra G., Nishida K. and Nagahara H. *Exploring the Latent Structure of AMD Severity Using Retinal Foundation Models*, IEEE EMBS International Conference on Biomedical and Health Informatics (Hong Kong **2026**), arXiv.2610.14138 [Accepted].

ACADEMIC SERVICE

Program Committee Member
2026
Siem Reap (Cambodia)

MultiMedia Modeling 2027 conference
Special Session: Multimodal Computer Vision in Medicine

INTERESTS

- **Longitudinal** modeling
- **Representation** learning
- **Multimodality**
- **Explainable** AI

OTHER EXPERIENCE

Internship

Data acquisition and analysis at Quality Control Department on: *Joint Commission* accreditation, Emergency Room flows, customer satisfaction.

Oct 2022 - Dec 2022

Gradenigo Hospital (Italy)

ACADEMIC PROJECTS

Deep learning:

- **Cardiosphere cell segmentation** from *microscopy images*.
  Full image processing pipeline (with pre- and post-processing) using *MMSegmentation* and *Detectron2* models. ~ *Python*
- **Backbone segmentation** from *CT-scan images*.
  Full image processing pipeline (with pre- and post-processing) using *MMSegmentation*. ~ *Python*

Machine learning:

- **Rectal cancer CAD systems** from *T2 MRI/ADC images*.
  Tumor segmentation and treatment response prediction using *Radiomics*-based classifiers. ~ *MATLAB*
- **Prediction of ventilation weaning** using *clinical data*.
  Multiple classifiers and *fuzzy logic* systems for prediction of successful mechanical ventilation weaning. ~ *MATLAB*
- **Gait status prediction** from lower limbs *EMG signals*.
  Walk/run detection using muscle synergies extracted from EMG signals, supported by statistical analysis. ~ *MATLAB*
- **Low-stakes lie detection** from *face video acquisitions*.
  Lie detection app with ensemble prediction using *MediaPipe* and *ARKit Blendshapes* for explainability visualization. ~ *Python, Unity*

Additional projects:

- **Depression detection system** from *video, audio and heart data*.
  Design of a hybrid deep / machine learning multimodal system for depression assessment using video speech acquisitions and heart data.
- **Baby growth telemonitor** platform from *body measurements*.
  Front and back-end development of a telemonitoring website for inputting and trend visualization of infants' body measurements. ~ *JS, HTML*
- **Pipeline software design** for heart telemonitoring platform.
  UML process design and user interface modeling for a heart failure treatment telemonitoring platform. ~ *UML*

Transversal skills

- **Interfacing** with interdisciplinary experts and researchers.
- **Adaptiveness** in international research environments.
- **Presenting** scientific results in written, oral and visual form.

PROGRAMMING

- Python
  PyTorch • *scikit-learn* • *TensorFlow* • *torchdiffeq* • *WebDataset* • *Mediapipe*
- MATLAB
- R
- HTML

SOFTWARE

Development

- Git
- Google Colab
- Kaggle
- Linux

Documents and graphics

- LaTeX
- 3D Slicer
- Unity
- SolidWorks

DATA ANALYSIS

Modalities

- Medical **imaging**
- Clinical **data**
- Biomedical **signals**

Techniques

- **Deep** learning
- **Machine** learning
- **Statistics**
- **Explainable** AI
- **Uncertainty** quantification

LANGUAGES

Italian *C2** ●●●●●● *Native*

English *C1** ●●●●●○ *IELTS Academic Band 8.0 (2022)*

Japanese *A1** ●○○○○○

*Using CEFR rating for proficiency.